

A CASE STUDY **ON** **ZERODHA**

By
Team Ojasvi

TEAM MEMBERS:

Mohd Aatif Sayyed
Abdulharis Shaikh
Ojasvi Mohapatra
Mohamed Reefath

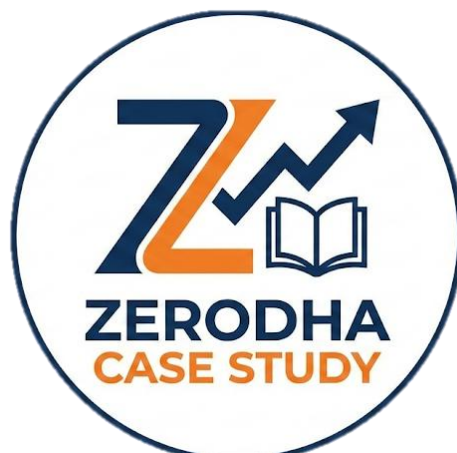


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1. Introduction

i. About Zerodha:

- Zerodha is one of the largest stock brokerage companies in India, founded in 2010. It serves millions of retail investors through platforms like Kite, Coin, and Console
- The company operates on a low-margin, high-volume business model. This approach depends heavily on efficient and scalable technology.
- Technology is the backbone of Zerodha's operations, ensuring speed, reliability, and trust. It enables the platform to handle millions of trades every day without disruption.
- Zerodha handles a highly time-sensitive trading environment. Even small delays can impact trade execution and user confidence.
- The platform experiences heavy traffic during market hours. Its systems are designed to handle sudden spikes efficiently and securely.

ii. Nature of the business:

- **Zerodha operates in a highly time-sensitive trading environment**, where even a small delay can directly affect user trades. During market opening and closing hours, there is a sudden surge in user activity and thousands of orders are placed simultaneously.
- **Trading volumes on the platform fluctuate heavily throughout the day.** The system must process all orders accurately and instantly, even during peak demand.
- **A fixed infrastructure is not suitable for such unpredictable workloads.** It may either fail during peak hours or remain underutilized during low-traffic periods.
- **To handle this bursty and unpredictable traffic efficiently**, Zerodha uses a hybrid infrastructure.
This combines on-premises systems for critical trading operations with cloud resources for scalability and cost efficiency.

iii. Why technology is critical for Zerodha:

- **For a trading platform like Zerodha**, technology directly affects speed, reliability, and customer trust. Any system downtime during trading hours can prevent users from entering or exiting trades, leading to financial losses.
- **To avoid such risks, Zerodha focuses strongly on system reliability.** Continuous monitoring, backup systems, and fail-safe mechanisms are used to ensure uninterrupted operations.
- **Zerodha uses cloud services such as AWS for flexibility and scalability.** This allows the platform to handle varying user loads efficiently.

- At the same time, core trading systems are kept on physical on-premise servers. This provides better control, lower latency, and improved performance for critical trading operations.
- **This hybrid approach helps Zerodha balance performance, reliability, and cost efficiency.** It also avoids dependence on a single infrastructure system.

2. Executive summary

- **Zerodha is one of the leading online stock trading platforms in India**, serving millions of retail customers. Its business operates in a highly time-sensitive environment where speed, uptime, and reliability directly impact customer trust.
- **To manage this scale efficiently, Zerodha uses a hybrid infrastructure model.** Core trading systems are hosted on on-premise servers with direct connectivity to stock exchanges to ensure low latency and high control.
- **Zerodha uses Amazon Web Services (AWS) for scalable and flexible workloads.** AWS supports customer-facing applications, background processing, and traffic-based resource scaling.
- **This combination of on-premise and cloud infrastructure balances performance, reliability, and cost efficiency.** Cloud features such as auto-scaling, batch processing, and storage help handle fluctuating user demand and large data volumes.

3. Problem Statement

i. Why traditional infrastructure is not enough:

Fixed, on-premises infrastructure cannot easily handle sudden increases in user traffic during market hours and leads to either underutilization or system overload.

ii. Risks of downtime and latency:

Even a small system outage or delay can prevent users from placing trades on time, resulting in financial loss and reduced customer trust.

iii. Difficulty in scaling during peak demand:

Trading activity spikes sharply during market open and volatile market conditions, making it difficult for traditional systems to scale quickly and efficiently.

4. Business Challenges

i. High number of users during market hours:

Zerodha experiences a large number of users logging in and placing trades simultaneously during market open and close, which puts heavy pressure on the system.

ii. Sudden traffic spikes:

Market volatility, news events, or IPO launches can cause unexpected surges in user activity, requiring the platform to handle sharp increases in traffic without slowing down.

iii. Need for fast and reliable trading:

Trading platforms must process orders with minimal delay, as even small latency issues can affect trade execution and user confidence.

iv. Regulatory and safety requirements:

As a regulated financial entity, Zerodha must follow strict compliance, data security, and system reliability standards set by regulatory authorities.

5. Hybrid Cloud Architecture and Technology Strategy

i. Why a hybrid approach was chosen:

Zerodha chose a hybrid approach to balance performance and flexibility. Critical trading operations require low latency and high control, while user-facing systems need scalability, which cannot be efficiently achieved using only one type of infrastructure.

ii. Combination of on-premises and cloud systems:

Core trading systems are maintained on on-premise infrastructure for direct exchange connectivity, while cloud services are used to handle user applications, background processing, and variable workloads.

iii. Key goals: speed, reliability, and cost control:

The primary objectives of this strategy are to ensure fast trade execution, maintain high system availability during peak demand, and optimize costs by scaling resources only when required.

5.1 On-Premises Infrastructure (Core Trading Systems)

i) Order Management System (OMS):

The Order Management System is the most critical component of Zerodha's trading platform. It is responsible for receiving buy and sell orders from users, validating them, and forwarding them to the stock exchanges for execution. Since trading decisions are extremely time-sensitive, the OMS must process orders with very low latency and high accuracy.

ii) Direct connectivity to stock exchanges:

Zerodha maintains direct and dedicated network connections with stock exchanges such as NSE and BSE. This direct connectivity helps reduce network delays and ensures faster order execution compared to routing orders through third-party systems.

iii) Reasons for keeping trading on-prem:

Core trading systems are kept on on-premises infrastructure to maintain full control, minimize latency, and ensure predictable performance. Hosting these systems in the cloud could introduce network delays and dependency risks, which are unacceptable for real-time trading operations.

5.2 Cloud Adoption Using AWS

i) Why AWS was selected:

Amazon Web Services was selected due to its reliability, scalability, global presence, and wide range of managed services. AWS allows Zerodha to scale infrastructure quickly without large upfront investments.

ii) Role of AWS in handling user traffic:

AWS is primarily used to manage customer-facing applications such as web and mobile platforms, APIs, background jobs, and data processing systems. During peak trading hours, AWS helps absorb sudden traffic spikes without impacting system performance.

iii) Benefits of cloud scalability:

AWS enables Zerodha to automatically scale computing resources based on real-time demand. This ensures high availability during peak hours while reducing costs during non-trading hours by scaling down unused resources.

5.3 Application & System Architecture

The Zerodha system follows a **hybrid architecture** that combines on-premise data centres with public cloud infrastructure (AWS) to achieve low latency, high availability, and scalability.

On-premise data centres host Zerodha's **core trading systems**, including the Order Management System (OMS). These systems are directly connected to stock exchanges through leased lines, which ensures fast trade execution and provides Zerodha with full control over system performance, reliability, and regulatory compliance.

The **public cloud (AWS)** is primarily used for customer-facing applications and workloads that require scalability. AWS supports various services such as compute (EC2, AWS Batch, and Lambda), storage (Amazon S3), networking (load balancers, Auto Scaling, and Multi-Availability Zone deployments), caching (Redis), and user notifications (Amazon Pinpoint). This cloud setup allows Zerodha to efficiently manage sudden traffic spikes during peak market hours while optimizing infrastructure costs.

CI/CD and orchestration are implemented using DevOps tools such as Terraform, Packer, eksctl, Kubernetes (EKS), and GitLab CI/CD. These tools automate infrastructure provisioning, testing, and application deployment, enabling faster releases, consistent environments, and reduced chances of manual errors.

Monitoring and logging are handled using Prometheus, Grafana, and the ELK stack. These tools provide real-time visibility into system health, performance metrics, and logs, allowing quick detection and resolution of issues during high-load or failure scenarios.

Backup and redundancy follow a multi-layered strategy. This includes AWS Multi-AZ availability, external cloud backups such as Azure-based systems, and communication-based alert mechanisms to maintain system awareness and continuity during failures.

Finally, **workload separation** is clearly defined within the architecture. Critical trading operations remain on-premise to ensure speed and control, while non-critical workloads such as analytics, batch processing, notifications, and monitoring are handled on the cloud. This separation improves overall performance, scalability, and cost efficiency.

5.4 AWS Services Used by Zerodha

1. Amazon EC2 (Elastic Compute Cloud)

Zerodha uses Amazon EC2 as the primary compute layer for running its user-facing applications and backend services. EC2 allows Zerodha to run application servers that power platforms like Kite and Console while giving full control over the operating system and runtime environment. This is important for Zerodha because it prefers predictable performance and the

flexibility to tune systems according to trading workloads, rather than relying entirely on abstracted managed services.

2. AWS Auto Scaling

Auto Scaling is used by Zerodha to automatically increase or decrease the number of EC2 instances based on real-time traffic. During market open, high volatility, or special events, user traffic can spike suddenly. Auto Scaling ensures that additional servers are provisioned automatically to handle the load, and scaled down during low-traffic periods, helping Zerodha maintain performance while controlling infrastructure costs.

3. Elastic Load Balancer (ELB / ALB)

Zerodha uses AWS load balancers to distribute incoming user traffic across multiple application servers. This prevents any single server from becoming a bottleneck and improves system reliability. Load balancing also plays a key role in high availability, ensuring that if one instance fails, traffic is automatically redirected to healthy instances without user impact.

4. Amazon S3 (Simple Storage Service)

Amazon S3 is used by Zerodha for storing logs, reports, historical data files, and other non-transactional data. S3 provides highly durable and cost-effective storage, making it suitable for large volumes of data that do not require real-time access. By using S3, Zerodha avoids maintaining large storage systems on-premise while ensuring data durability and easy access for analytics and audits.

5. AWS Batch

AWS Batch is one of the most clearly documented AWS services used by Zerodha. It is used to run large-scale background processing jobs such as profit-and-loss calculations and data processing tasks. Zerodha adopted AWS Batch to significantly reduce processing time by running jobs in parallel across multiple compute instances. This allows heavy workloads to be completed faster without affecting live trading systems.

6. AWS Lambda (Limited Use)

Zerodha uses AWS Lambda only for specific lightweight or event-driven tasks, not for core trading workflows. Lambda is suitable for small background operations where servers do not need to be running continuously. However, Zerodha intentionally avoids using Lambda for latency-sensitive or critical systems, preferring more controlled environments like EC2 for predictable performance.

7. Amazon EKS (Elastic Kubernetes Service)

Zerodha uses Kubernetes on AWS (EKS) to manage containerized applications and standardize deployments. Kubernetes helps Zerodha automate application scaling, manage service isolation, and deploy updates reliably. This aligns with Zerodha's infrastructure-as-code and automation philosophy, enabling consistent environments across development and production.

8. AWS Networking (VPC, Private Connectivity)

Zerodha uses AWS Virtual Private Cloud (VPC) and private networking to securely connect cloud infrastructure with on-premise data centers. This private connection allows data to flow securely between AWS and core trading systems without relying on the public internet. It helps maintain low latency and strengthens security for sensitive operations.

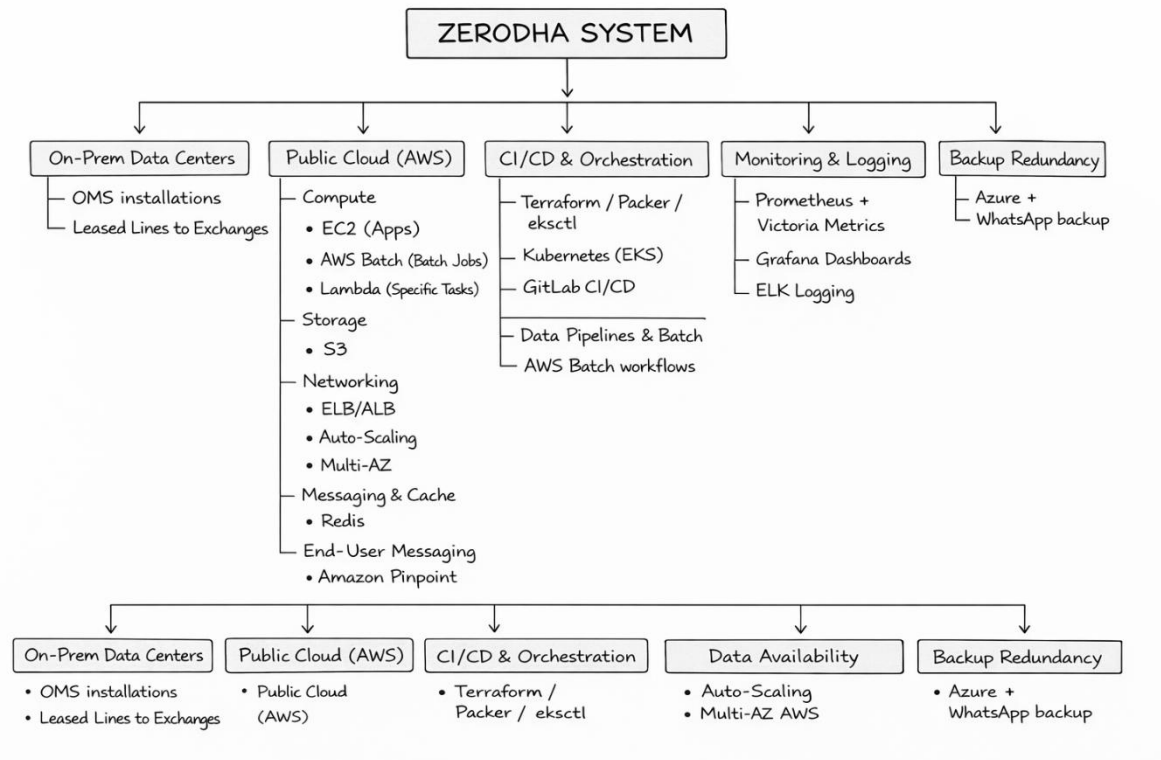
9. Monitoring: Limited Use of CloudWatch

Zerodha does not rely heavily on AWS CloudWatch for monitoring. Instead, it primarily uses open-source tools such as Prometheus, Grafana, and ELK for observability across both cloud and on-premise environments. CloudWatch may be used at a basic level, but Zerodha prefers unified monitoring tools that work consistently across hybrid infrastructure.

10. Services Zerodha Intentionally Avoids

Zerodha has publicly stated that it does not heavily depend on managed AWS database services such as Amazon DynamoDB or Amazon RDS for core systems. Instead, it prefers self-managed databases and infrastructure where it has greater visibility, control, and cost predictability. This choice aligns with Zerodha's engineering philosophy of avoiding unnecessary abstraction in latency-sensitive financial systems.

Zerodha Cloud Architecture Using AWS Infrastructure



High-Level System Architecture of Zerodha (Hybrid On-Premise and Cloud Model)

6. Key Learnings

i) Importance of hybrid architecture:

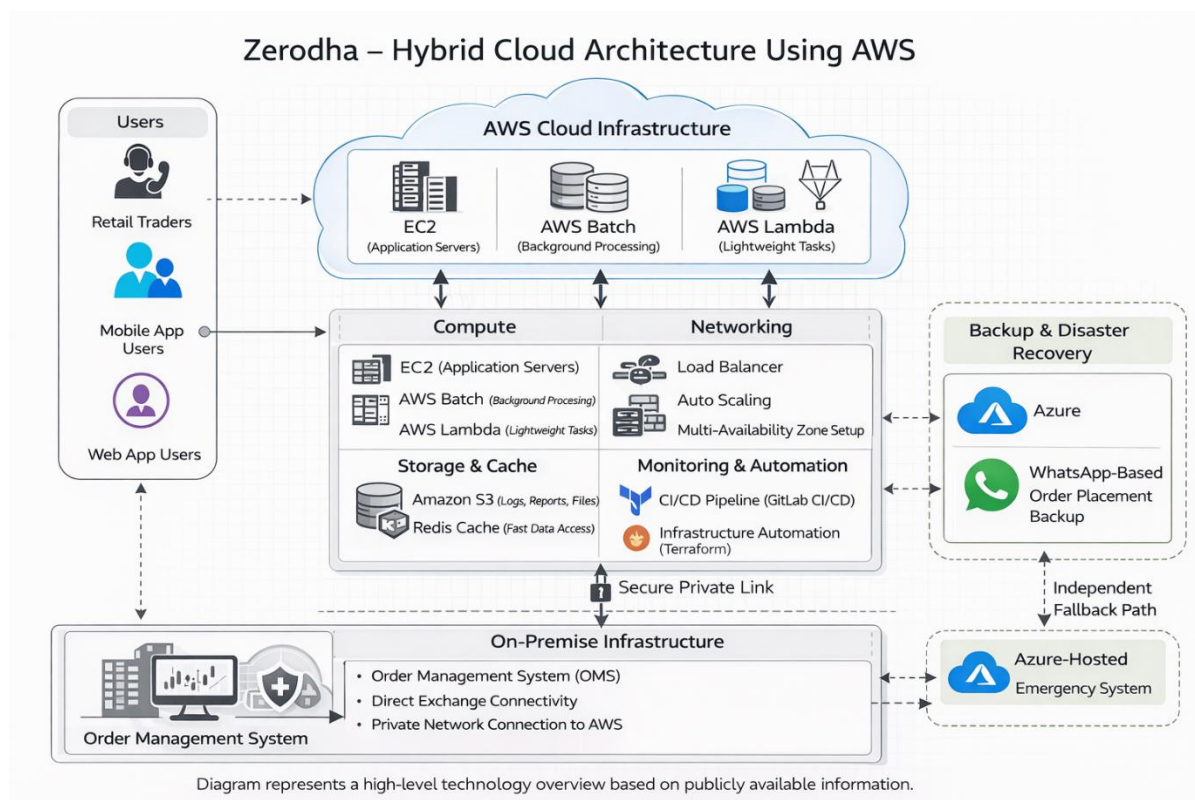
A hybrid architecture helps combine the speed and control of on-premises systems with the scalability and flexibility of cloud services. This approach is especially useful for handling time-sensitive trading operations.

ii) Avoiding single points of failure:

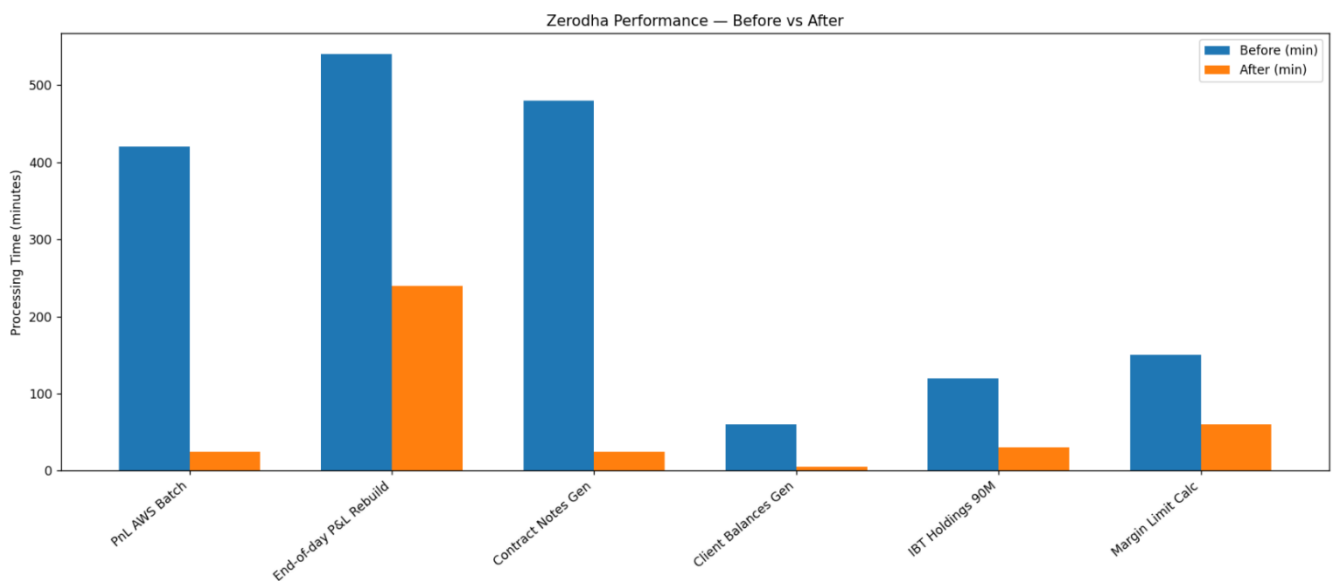
Designing systems with redundancy ensures that failure of one component does not bring down the entire platform. This improves system availability and reliability during peak trading hours.

iii) Balancing cost and performance:

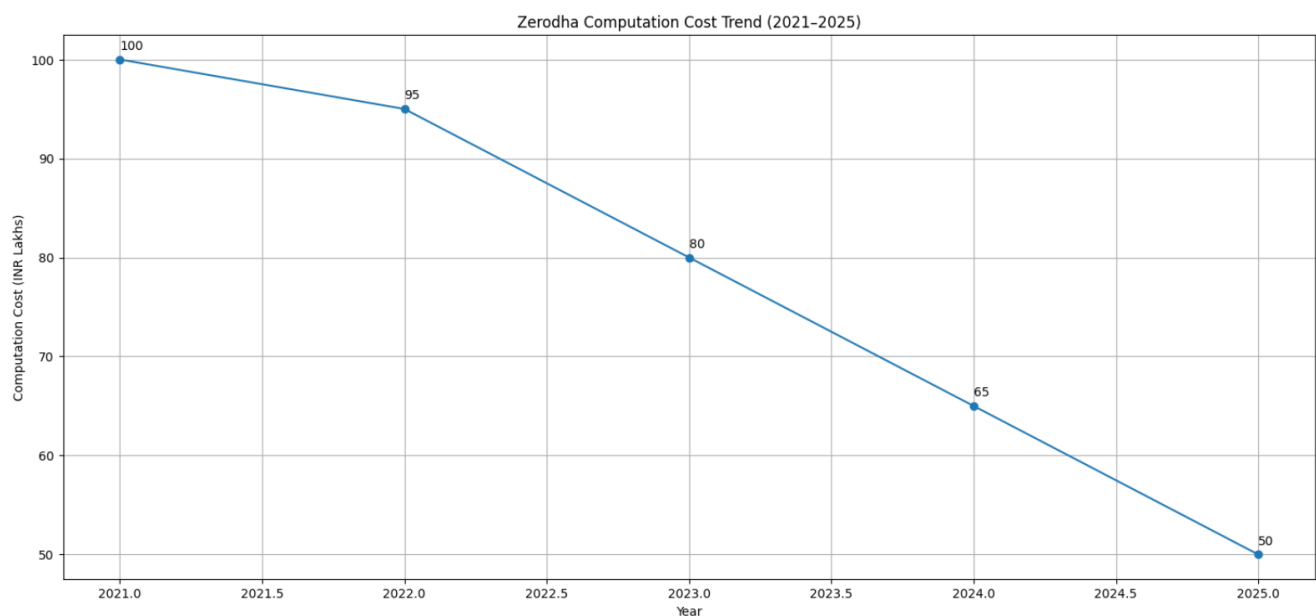
Using cloud resources only when needed allows the platform to maintain high performance during peak loads while reducing infrastructure costs during low-traffic periods.



7. Statistics



Performance graph showing before and after changes in processing time



Change in cost of computing over the years 2022-2025

8. Future Scope

i) Further automation:

Increasing automation in deployment, monitoring, and infrastructure management to reduce manual effort and human errors. By improving automation through CI/CD pipelines and infrastructure-as-code practices, system updates can be deployed faster and more consistently. This also helps in quicker recovery from failures and ensures that infrastructure changes are predictable and repeatable across environments.

ii) Improved resilience:

Strengthening backup, failover, and disaster recovery mechanisms to ensure system availability during failures or high-load conditions. This includes improving redundancy across multiple zones or regions and regularly testing failover processes. A stronger resilience strategy helps minimize downtime and ensures that critical trading services remain accessible even during unexpected disruptions.

iii) Scaling for future growth:

Expanding infrastructure capabilities to support a growing user base, higher trading volumes, and new features efficiently. As the number of users increases, the system must handle higher concurrency without performance degradation. Scalable cloud resources and optimized system design will allow Zerodha to introduce new products and features while maintaining stability and cost efficiency.

9. Conclusion

Zerodha's cloud approach combines on-premises infrastructure with cloud services to achieve low latency, high reliability, and scalability. Zerodha's technology setup is built around a clear and practical hybrid approach. Core trading operations are maintained on on-premises infrastructure to ensure low latency, control, and regulatory compliance, while cloud services are used to support user-facing applications and scalable workloads. This separation allows Zerodha to handle high trading volumes efficiently without compromising performance or reliability.

By using AWS for scalability, automation, monitoring, and background processing, Zerodha can manage sudden traffic spikes during market hours while keeping costs under control. Continuous monitoring, automated deployments, and an independent backup system further strengthen platform stability. Overall, Zerodha's cloud strategy demonstrates how a carefully designed hybrid model can effectively support the demands of a large-scale fintech platform.

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